COURSE OUTLINE

Department/ Faculty:	Software Engineering/Computing	**Page:**	1 of 8
Course code:	SECJ 3483	**Academic Session/Semester:**	2024/2025/Sem II
Course name:	Web Technology	**Pre/co requisite (course name and code, if applicable):**	Web Programming (PHP) Object Oriented Prog
Credit hours:	3		

Course synopsis	In this course, students will learn web development techniques, focusing on modern frontend and backend technologies for building full-stack web applications. Topics include Vue.js for frontend development, PHP Slim as a lightweight backend framework, MySQL for database management, and cloud deployment for real-world applications. Students will gain hands-on experience in designing, developing, and deploying scalable web applications. By the end of the course, students will be able to build and deploy functional web applications with a complete understanding of both frontend and backend integration.			
Course coordinator	Norizam Katmon			
Course lecturer(s)	**Name**	**Office**	**Tel No/WA**	**E-mail (@utm.my)**
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Prepared by:

Name:	(Course Owner)
Signature:	
Date:	24 August 2017

Certified by:

Name:	PM. Dr.Radziah Mohamed (Director of Software Engineering Dept.)
Signature:	
Date:	

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Mapping of the Course Learning Outcomes to the Programme Learning Outcomes, Teaching & Learning (T&L) methods and Assessment methods:

No.	Course Learning Outcome	*Program Learning Outcome	Weight (%)	**Taxonomies and generic skills	T&L methods	***Assessment methods
CLO1 (Knowledge) Individual	Explain the principles, architecture, and key technologies of modern web development, including frontend, backend, databases, APIs, security and deployment.	PO1 (Knowledge Understanding)	25	C2 Understanding	Lecture, Active Learning (Readings, Discussions, Online Tutorials)	Quiz, Test1, Final
CLO2 (Skills) Individual	Develop a functional and optimized full-stack web application individually using related web technologies	PO2 (Problem Solving)	55	P3 Application	Hands-on Labs, Individual Coding Assignments, Debugging, Exercises	Test1, Assg, Lab Exercise, Final
CLO3 (Skills) Group Work)	Collaborate within a team to build a full-stack web application using modern web technologies	PO3 (Technical Skill)	15	P4 Integration	Group Project, Code Reviews, Team-Based Learning	Project (Product Features & Completeness)
CLO4 (Attitude) Group Work & Ethics	Demonstrate teamwork, ethical considerations, and professional responsibility in web development projects, ensuring secure coding practices and effective collaboration within a group setting.	PO8 (Team Working)	5	A3, TS2 Teamwork & Ethics	Project Based Learning (Group Discussions, Project Collaboration, Peer Reviews)	Project (Presentation & Teamwork)

*Program Learning Outcome
PO1 - Ability to acquire and apply theory and principles of computer science and equip with social science and personal development knowledge.
PO2 - Ability to design and construct computer programs using standard approaches
PO3 - Ability to design and construct computer programs using standard approaches.
PO8 - Ability to work effectively in a team

**Taxonomies and Generic Skills

*** LE – Lab exercise, Q – Quiz, A – Assignment, T1 – Test 1,, F – Final Exam, PR – Mini Project, Demo– Presentation
L – Lecture, T – Tutorial, P – Practical

Details on Innovative T&L practices:

No.	Type	Implementation

1.	Active learning	Conducted through in-class activities
2.	Independent Study	E-learning
3.	Project-based learning	Conducted through study assignments and mini project. Tasks are given in sequential steps throughout the semester. Students in a group of 3-4 are required to develop both the front-end view and back-end API development server for the selected case study.

Weekly Schedule:

Pertemuan 1	**Introduction to Web Technology** Overview of web technologies Frontend, Backend, Database, API, Deployment. Web application architecture. Setting up the development tools (VS Code + Extensions, Postman, XAMPP-MySql, GitHub).
	Front-End Development Fundamentals HTML & CSS Review (Structure, Styling, Box Model, Responsive Design). Introduction to DOM, AJAX, JSON & XML for dynamic content loading. JavaScript Basics (Syntax, Variables, Functions). DOM Manipulation & Event Handling.

Pertemuan 2	**JavaScript and HTML DOM** (Quiz 1) Basics of JavaScript: Syntax, Variables, and Functions HTML Document Object Model (DOM) Introduction to HTML DOM Manipulation Event Handling in JavaScript JavaScript Control over HTML DOM
	jQuery and AJAX Integration (Lab 1) Why Use jQuery? (Handling Legacy Code) DOM Manipulation using jQuery ($("selector")). AJAX with jQuery vs Fetch API. Making API Calls with jQuery AJAX. **Group formation**
Pertemuan 3	**Asynchronous Programming in JavaScript** Javascript Async: Concepts and Benefits Async Implementations: XmlHttpRequest (HXR), JQuery AJAX, Promise, Fetch, Async/Await
	Frontend Framework - Vue.js Introduction to Vue.js Setting up Vue.js (CDN & CLI) Vue Directives (v-bind, v-model, v-if, v-for) Vue Components (Props, Events & Emits)
Pertemuan 4	**Frontend Development with Vue.js** Building User Interfaces with Vue Components Vue Router for Navigation (SPA). Vue State Management using Composition API
	Frontend Development with Vue.js (cont…) (Lab 2) Fetching & Displaying Data from APIs. Handling Forms & Validation Integrating Vue.js with a Mock API (JSON Server). **Test1 (written test)** **(Date: ??? May 2025 - ??? . Venue : ???)**
Pertemuan 5	**Back-end Development with PHP Slim** (Assg 2) RESTful APIs using PHP Slim Routing and Middleware in PHP Slim Creating RESTful API Endpoints (CRUD with MySQL). **Group Project Presentation - Progress (Front-end using Vue)**
	Back-end Development with PHP Slim (cont…)(Lab 3) Building RESTful APIs using PHP Slim and MySQL Database Implementing CRUD Operations (Create, Read, Update, Delete)

Pertemuan 6	**Securing APIs with JWT Authentication** Understanding Token-Based Authentication using JWT. Implementing JWT Login & Token Validation. Protecting API Routes with Middleware.
	Back-end Security Understanding Common Web Security Threats Cross-Site Scripting, SQL Injection Query Data Tampering Authentication and Authorization
Pertemuan 7	**Full-Stack Integration (Vue + PHP Slim + MySQL)** Connecting Vue frontend with PHP Slim backend. Handling API requests, authentication & data storage. **Cloud Deployment** Deploying Full-Stack Web App on DigitalOcean Droplets (Linux + Nginx + MySQL + PHP).
	Group Project Demo/Presentations
Week 16	**REVISION WEEK (1 WEEK)**
Weeks 17 - 19	**FINAL EXAM WEEKS**

Transferable skills (generic skills learned in course of study which can be useful and utilised in other settings):

Design both the frontend and backend web application individually and in team
Develop both the frontend and backend web application individually and in team
Team working and Presentation

Student learning time (SLT) details:

Distribution of student Learning Time (SLT) Course content outline					Teaching and Learning Activities		TOTAL SLT
	Guided Learning (Face to Face)				Guided Learning Non-Face to Face	Independent Learning Non-Face to face	
CLO	**L**	**T**	**P**	**O**			
CLO 1	21h	15h	9h		3h	15h	**63h**
CLO 2	11h	9h	13h		3h	10h	**46h**
CLO 3	2h		8h		3h	5h	**18h**
Total SLT	**34h**	**24h**	**30h**		**9h**	**30h**	**127h**

Assessment Plan

No	Assessment	Frequency	% Total
1	Quizzes	2	10
2	Labs	3	10
3	Assignments	2	10
4	Test 1	1	20
5	Group Project Presentation & Demo	1 1	15 5
6	Final Exam (Written)	1	30
	TOTAL		**100**

Special requirement to deliver the course (e.g: software, nursery, computer lab, simulation room):

Software:

Visual Studio Code, JQuery, Xampp, MySQL, PHP Slim, Vue.js.

Learning Resources:

Main reference:

Chris Aquino, Todd Gandee (2016). Front-End Web Development : The Big Nerd Ranch Guide. ISBN10 0134433947

Additional references:

Oswald Campesato, Kevin Nilson (2010).WEB 2.0 Fundamentals. Jones and Barlett. ISBN 978-0- 7637-7973-3

Nathan Murray, Ari Lerner, Felipe Coury, Carlos Taborda (2017). ng-book: The Complete Guide to Angular 4. ISBN-10: 1546376232

John M. Wargo (2015). Apache Cordova 4 Programming (Mobile Programming) 1st Edition. ISBN-10: 0134048199

Course Website:

Utm E-learning at "http://elearning.utm.my/"

Academic honesty Learning resources and plagiarism:

Copying of work (texts, lab results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of **zero** for the assignment and exams and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (Course policy, any specific instruction etc.):

1. Attendance is compulsory and will be taken in every lecture session. Student with less than 80% of total attendance is not allowed to sit for final exam.
2. Students are required to behave and follow the University's dressing regulation and etiquette all the timel.
3. Exercises and tutorial will be given in class and some may be taken for assessment. Students who do not do the exercise will lose the coursework marks for the exercise.
4. Assignments must be submitted on the due dates. Some points will be deducted for late submissions. Assignments submitted **three days** after the due date will not be accepted.

Make up exam will not be given, except to students who are sick and submit medical certificate confirmed by UTM panel doctors. Make up exam can only be given within one week of the initial date of exam.

Disclaimer:

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